

Harmankaranjit Singh

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EDUCATION

UNIVERSITY OF MARYLAND BALTIMORE COUNTY

Baltimore, USA

Master of Science in Data Science

- **Relevant Coursework:** Data Science, Machine Learning, Big Data Analytics, Statistics, and Visualization

GANDHI INSTITUTE OF TECHNOLOGY AND MANAGEMENT

Hyderabad, India

Bachelor of Technology in Computer Science Engineering

- **Relevant Coursework:** Python Programming, Database Management System, SDE, Object-Oriented Programming

TECHNICAL SKILLS

Programming Languages : Python, SQL, R, Java, Javascript, HTML, CSS, C, C++.

Cloud Technologies : AWS, Azure.

ML and AI Tools : TensorFlow, Pytorch, Pandas, Numpy, Scikit-learn, Keras, MLflow, Seaborne, OpenCV, AWS Sagemaker.

Databases : MySQL, Postgre SQL, AWS RDS, S3, Redshift, Athena, Elasticsearch, MongoDB.

ETL/Data Warehouse Tools : AWS Glue, Data Migration, Informatica, Snowflake, Airflow, Salesforce.

Visualization : Power BI, Tableau, AWS Quicksight.

Web Application & Tools : Streamlit, Dash, Microsoft Excel.

PROFESSIONAL EXPERIENCE

VITG

BALTIMORE, USA

Data Analyst

June 2023- Present

- Developed predictive models using machine learning frameworks in AWS Sagemaker, Python (Scikit-learn, TensorFlow).
- Collaborated with Cross Functional Teams to integrate AI-driven insights into dashboards, reports with PowerBi & Tableau.
- Designed & Integrated scalable ETL/ELT pipelines for advanced analytics, Streamlining & leveraging cloud platforms like AWS.
- Developed Power BI dashboards for KPIs and business metrics, providing actionable insights for Business Stakeholders.
- Analyzed large datasets using SQL, Python, and Excel, driving operational optimization and uncovering data-driven insights.
- Partnered with leadership to define the organization's data strategy & explored opportunities for AI adoption.
- Integrated SQL, NoSQL databases and AWS RDS with cloud storage, enabling real-time data access and seamless analytics.
- Improved report (EHR) accuracy by 35% through enhanced EDA, Power BI features, and optimized workflows in Redshift.
- Integrated Salesforce with cloud and data warehouse solutions, enhancing CRM analytics and streamlining data processes.

UNIVERSITY OF MARYLAND BALTIMORE COUNTY

BALTIMORE, USA

Data Analyst (Graduate Assistant)

Aug 2022 – Jun 2023

- Developed and deployed ML models on AWS Sagemaker, automating patient condition analysis and improving model efficiency.
- Extracted, cleaned, and analyzed patient data using SQL and Python, enhancing data-driven insights for diagnostics.
- Automated chest X-ray analysis using CNNs with TensorFlow, achieving 95% accuracy and reducing diagnostic time by 50%.
- Improved MRI resolution by 30% using Python, OpenCV, and KNN interpolation, delivering clearer diagnostic images.
- Deployed predictive models on AWS EC2, enabling real-time COVID-19 predictions and improving patient management.
- Built ML models with Random Forests and Gradient Boosting, achieving 90% accuracy in predictions.
- Streamlined data workflows using Python and AWS Cloud, enhancing efficiency in healthcare model training and analysis.
- Presented findings through interactive Tableau visualizations, supporting healthcare professionals in making accurate diagnoses.

DHARMA SOLUTIONS

HYDERABAD, INDIA

Data Analyst

May 2021 – Aug 2022

- Implemented automated error detection tools with Python and SQL, leading to 30% reduction in manual review times.
- Performed statistical analysis using R and Python, lowering Human Machine Interface (HMI) false positives by 18%.
- Led a team to optimize workflow with data validation protocols, boosting processing speed by 25%.
- Managed projects with Agile methodologies, optimizing resource allocation to improve project throughput by 20%.
- Streamlined reporting processes by developing interactive dashboards using Tableau, increasing data accuracy by 15%.
- Analyzed business requirements and implemented new CRM systems, resulting in a 20% increase in customer satisfaction.
- Developed actuarial models to assess financial risks, forecast claims trends, and optimize pricing strategies using Python, R, and SQL.
- Improved data quality for ML models with pandas and NumPy, enhancing performance.

DEFENCE RESEARCH AND DEVELOPMENT ORGANIZATION (DRDO)

HYDERABAD, INDIA

Data Analyst

May 2019 - July 2019

- Engineered Python solutions, utilizing NumPy, Matplotlib and Scikit-learn, to detect patterns in high-frequency data.
- Performed data cleaning and validation using Python, SQL and Excel ensuring data accuracy and consistency.
- Developed basic dashboards and reports in Tableau and Power BI, supporting internal team needs.
- Conducted ad-hoc analysis, extracting actionable insights from structured datasets to guide decision-making.
- Maintained comprehensive documentation for data workflows, driving process improvements and efficiency.

CERTIFICATIONS AND TRAINING

- AWS Certified AI Practitioner.
- AWS Certified Cloud Practitioner.
- Tableau Desktop Specialist.
- Microsoft Power BI Data Analyst.

